REPORT

OF

DATA SCIENCE MASTER VIRTUAL INTERNSHIP

Submitted To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech Computer Science and Engineering

By

KUSHAGRA PRATAP SINGH

Roll No.: 2410030890

Section: 2CSE15

Batch: 2024-28

1. CANDIDATE'S DECLARATION

I, Kushagra Pratap Singh, **do hereby declare** that the internship report titled "Data Science Master Virtual Internship" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering at IILM University, Greater Noida, U.P.

I further declare that this report presents my work and learning experience during the eight-week virtual internship conducted from June 2026 to August 2026. All references and source materials used in preparing this report have been acknowledged appropriately. I have not knowingly reproduced material from another source as my own.

I affirm that this report has not been submitted to any other institute or university for the fulfilment of any degree or diploma requirement. I understand that I am responsible for ensuring that the report complies with applicable copyright and academic-integrity requirements, and I shall be solely responsible for any copyright violation arising from the submission.

Signature: ______________________________

Student Name: Kushagra Pratap Singh

Roll No.: 2410030890

2. ACKNOWLEDGMENT

I take this opportunity to express my sincere gratitude to all the individuals and organizations whose guidance, support, and encouragement contributed to the successful completion of my Data Science Master Virtual Internship.

I am grateful to the All India Council for Technical Education (AICTE), Ministry of Education, for its collaboration and support in enabling the internship opportunity. I also express my sincere appreciation to EduSkills for organizing and facilitating the virtual internship and for providing a structured learning pathway covering applications and use cases, data engineering, machine learning, data preparation, deployment, monitoring, and platform administration.

I extend my thanks to Siemens Industry Software (India) Pvt. Ltd. for supporting the internship. I am especially thankful for the professional learning framework represented by the internship and for the opportunity to engage with concepts and practices relevant to data science and enterprise AI environments.

I would also like to acknowledge the contribution of the key people associated with the internship: Mr. Gaurav Pai, Head, Strategy and Operations, Siemens Industry Software (India) Pvt. Ltd.; Dr. Buddha Chandrasekhar, Chief Coordinating Officer (CCO), AICTE, Ministry of Education; and Mr. Shubhajit Jagadev, Chief Executive Officer (CEO), EduSkills.

I express my sincere gratitude to the faculty members and academic mentors of the School of Computer Science and Engineering, IILM University, Greater Noida, for their academic guidance and encouragement. Their support helped me connect the internship learning with my B.Tech Computer Science and Engineering studies.

Finally, I am deeply thankful to my parents for their blessings, encouragement, and continuous support throughout the internship and the preparation of this report.

Signature: ______________________________

Student Name: Kushagra Pratap Singh

Roll No.: 2410030890

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3. INTERNSHIP COMPLETION CERTIFICATE

शिक्षा मंत्रालय
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All India Council for Technical Education

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AICTE - EduSkills
2026
VIRTUAL INTERNSHIP

Certificate of Virtual Internship

This is to certify that

Kushagra Pratap Singh

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Science Master Virtual Internship

During June - August 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS

G.V. Pai
Gaurav Pai
Head - Strategy and Operations
Siemens Industry Software (India) Pvt. Ltd

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

Certificate ID : 4d1eb79183bf19bf6a43

Student ID: STU6a0dfd7bdb6161779301755

INTERNSHIP GRADE
O

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

4. PROJECT DESCRIPTION

4.1 Introduction

Data science is an interdisciplinary area concerned with the systematic use of data to understand patterns, generate insights, support decisions, and develop data-driven applications. In contemporary computing environments, data science combines data access and engineering, statistical reasoning, machine learning, visualization, model evaluation, and operational practices. The effectiveness of a data science solution therefore depends not only on the ability to build a model but also on the ability to understand a business problem, prepare reliable data, evaluate results, deploy a solution, and manage the platform on which analytical workflows operate.

The Data Science Master Virtual Internship was undertaken as an eight-week, unpaid, virtual learning experience during June 2026 to August 2026. It was organized by EduSkills, supported by Siemens Industry Software (India) Pvt. Ltd., and carried out in collaboration with AICTE, Ministry of Education. The internship was designed as a progressive learning pathway that moved from identifying suitable AI and machine-learning use cases to data engineering, machine learning, advanced data preparation, deployment and monitoring, and platform administration.

The internship is particularly relevant to a B.Tech Computer Science and Engineering student because academic study commonly introduces algorithms, programming, databases, statistics, artificial intelligence, and machine learning as individual subjects. An industry-oriented learning pathway provides an opportunity to connect these areas into a broader workflow. The present report documents the structure of the internship, its learning objectives, the tools and concepts involved, the methodology followed, and the expected professional outcomes.

The scope of the internship was broader than a single isolated machine-learning exercise. It covered the lifecycle of an analytical solution: understanding the problem, accessing and transforming data, preparing features and datasets, developing models, considering deployment and monitoring, and understanding the administration of an AI platform. This lifecycle perspective is important because real-world data science requires coordination between analytical reasoning and operational implementation.

4.2 Organization Profile

EduSkills

EduSkills was the organizing body for the internship. Within the internship structure provided for this report, EduSkills served as the learning and coordination platform through which the eight-week Data Science Master Virtual Internship was delivered. The program was designed to provide a structured sequence of professional and master-level modules rather than a single-topic training activity.

Mr. Shubhajit Jagadev, Chief Executive Officer (CEO), EduSkills, is identified on the internship certificate as one of the key officials associated with the program. His designation establishes the organizational leadership represented in the internship documentation.

Siemens Industry Software (India) Pvt. Ltd.

Siemens Industry Software (India) Pvt. Ltd. supported the Data Science Master Virtual Internship. The internship framework connected the learning experience with enterprise-oriented AI and data-science practices, including data preparation, machine learning, deployment, monitoring, and platform administration.

Mr. Gaurav Pai, Head, Strategy and Operations, Siemens Industry Software (India) Pvt. Ltd., is identified as a key person associated with the internship. The certificate issued for completion carries his designation and signature.

AICTE, Ministry of Education

The All India Council for Technical Education (AICTE), Ministry of Education, was associated with the internship in collaboration with EduSkills and Siemens. The internship documentation identifies an AICTE Student ID for the candidate and places the program within the stated collaboration framework.

Dr. Buddha Chandrasekhar, Chief Coordinating Officer (CCO), AICTE, Ministry of Education, is identified as a key official associated with the internship and appears on the completion certificate.

IILM University, Greater Noida

IILM University, Greater Noida, U.P., is the academic institution of the candidate. The internship learning is documented as part of the student's B.Tech Computer Science and Engineering academic journey under the School of Computer Science and Engineering. The report has been prepared in accordance with the provided internship-report format and academic submission requirements.

4.3 Problem Statement

A major challenge in undergraduate technical education is the transition from subject-wise academic learning to integrated industry practice. A student may study programming, databases, artificial intelligence, statistics, and machine learning, yet still require structured exposure to how these capabilities are combined around an actual business or organizational problem.

In data science, this gap is particularly visible because a complete solution involves more than selecting and training an algorithm. The practitioner must first identify a meaningful use case, understand what data is required, obtain and transform that data, prepare it for analysis, build and assess a model, and then consider deployment, monitoring, and the administration of the platform supporting the solution.

The internship therefore addressed the learning need for a progressive, end-to-end understanding of data-science practice. Its sequence of modules was intended to bridge foundational concepts and professional application by moving from applications and use cases through data engineering and machine learning, and subsequently into advanced preparation, deployment, monitoring, and platform administration.

The central problem addressed by the internship can consequently be stated as follows: how can a computer science student develop an integrated understanding of the data-science lifecycle and acquire practical exposure to tools and workflows that connect business problems, data, machine-learning models, and enterprise platform operations?

4.4 Project Objectives

- To understand how business and organizational problems can be evaluated for suitability for artificial intelligence and machine learning.

- To develop foundational competence in accessing, loading, transforming, and calculating data.

- To strengthen machine-learning knowledge through a structured progression from foundational to advanced concepts.

- To develop advanced data-preparation and automation awareness for reliable analytical workflows.

- To understand the transition from model development to deployment and monitoring.

- To gain exposure to RapidMiner AI Hub and concepts associated with AI-platform configuration and administration.

- To understand enterprise-level platform operations as part of the wider data-science lifecycle.

- To complete weekly learning and assessment activities in a disciplined virtual-learning environment.

- To validate internship completion and learning through the final assessment process.

- To improve readiness for future academic projects, internships, and entry-level data-science or machine-learning responsibilities.

4.5 Scope of the Project

The scope of the internship was defined by the eight-week module sequence supplied for the program. The sequence progressed from professional-level applications and data engineering to master-level data preparation, machine learning, deployment, and platform administration. The following table records the complete scope.

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Build foundational machine-learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine-learning

		techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
Week 7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.
Final	Credential Validation / Internship Grade Point Assessment	Final Assessment Test for validation and internship grade-point assessment.

Table 4.1 Week-wise internship structure

Week 1: Applications & Use Cases Professional

The first week established the problem-oriented perspective of data science. The emphasis was on recognizing situations in which artificial intelligence or machine learning can be meaningfully applied. This stage is important because a technically sophisticated model is useful only when it addresses a clearly defined need. The learning focus therefore connected business problems with the potential role of AI/ML.

Week 2: Data Engineering Professional

The second week focused on data engineering. Data must be accessible, correctly loaded, transformed into usable forms, and subjected to appropriate calculations before meaningful analytical work can begin. This module introduced the engineering side of the data-science workflow and emphasized the importance of preparing data in a systematic manner.

Week 3: Machine Learning Professional

The third week developed foundational machine-learning skills. After establishing the use-case and data-engineering context, the learning sequence moved to model-oriented thinking. The purpose was to understand the fundamental concepts required to build machine-learning solutions and to relate those concepts to the prepared data.

Week 4: Data Preparation Master

The fourth week advanced the data-preparation component through the Data Preparation Master module. The focus on advanced data handling and automation reflects the fact that practical analytics frequently requires repeatable and reliable preparation workflows rather than one-time manual transformations.

Week 5: Machine Learning Master

The fifth week extended the machine-learning foundation into advanced techniques. This stage strengthened the progression from basic model-building concepts toward more capable analytical workflows and a deeper understanding of machine-learning practice.

Week 6: Applications & Use Cases Master

The sixth week addressed the operational side of machine learning through deployment and monitoring. A model has practical value only when it can be made available for use and its behavior can be observed over time. This module therefore connected analytical development with application-oriented implementation.

Week 7: Platform Administration Master

The seventh week introduced platform administration through RapidMiner AI Hub. The focus shifted from individual analytical workflows to configuration and management of the platform environment that supports AI work.

Week 8: Platform Administration Master

The eighth week continued platform administration with an enterprise-level perspective. This provided a broader operational view of how an AI platform may be managed beyond the immediate construction of an individual model or workflow.

4.6 Technologies and Tools Used

Technology / Tool	Role in Internship
RapidMiner AI Hub	The internship included exposure to RapidMiner AI Hub, particularly in the platform-administration modules. It served as the named AI-platform environment for configuration and management learning.
Data engineering tools and workflows	The data-engineering component addressed accessing, loading, transforming, and calculating data. These operations form the foundation for building dependable analytical datasets.
Machine-learning tools and libraries	The machine-learning modules provided a structured environment for developing foundational and advanced machine-learning skills. Specific external library names were not supplied in the internship details and are therefore not attributed here.
Data preparation and automation capabilities	The Data Preparation Master module covered advanced data handling and automation, emphasizing repeatable preparation of analytical data.
Visualization and analytical presentation tools	Visualization was treated as a supporting capability for understanding and communicating analytical results. No specific visualization software was supplied in the internship details; accordingly, no

	specific product is claimed in this report.
Virtual learning platform	The internship was delivered in virtual mode, enabling structured access to learning modules and assessments without requiring physical attendance.

Table 4.2 Technologies and tools used / covered

4.7 System Architecture

The internship can be represented as a layered data-science architecture. The layers describe a logical progression rather than a claim about a particular software deployment. Each layer depends on the output of the preceding stage and contributes to the end-to-end lifecycle of a data-science solution.

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┌───────────────────────────────────────────────┐
│              BUSINESS USE CASES               │
│  Identify problems suitable for AI / ML       │
└───────────────────────┬───────────────────────┘
                        │
                        ▼
┌───────────────────────────────────────────────┐
│                DATA ENGINEERING               │
│  Access → Load → Transform → Calculate        │
└───────────────────────┬───────────────────────┘
                        │
                        ▼
┌───────────────────────────────────────────────┐
│                MACHINE LEARNING               │
│  Prepare → Train → Evaluate → Improve         │
└───────────────────────┬───────────────────────┘
                        │
                        ▼
┌───────────────────────────────────────────────┐
│            DEPLOYMENT & MONITORING            │
│  Operationalize models and observe behavior   │
└───────────────────────┬───────────────────────┘
                        │
                        ▼
┌───────────────────────────────────────────────┐
│            PLATFORM ADMINISTRATION            │
│  Configure → Manage → Enterprise Operations   │
└───────────────────────────────────────────────┘
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Figure 4.1 Layered system architecture for the internship learning pathway

Business Use Cases: The first layer establishes the reason for using AI or machine learning. The objective is to frame a problem clearly and determine whether a data-driven solution is appropriate.

Data Engineering: The second layer converts available data into forms suitable for analysis. Access, loading, transformation, and calculation are essential because model quality depends on the quality and suitability of the data supplied to the learning process.

Machine Learning: The third layer represents model development. Foundational and advanced machine-learning concepts are applied after the data has been prepared, with evaluation used to understand model behavior and guide improvement.

Deployment and Monitoring: The fourth layer represents the transition from an analytical model to an operational capability. Deployment makes a model available for use, while monitoring provides a means of observing its ongoing behavior.

Platform Administration: The final layer represents the environment in which AI workflows are configured and managed. The internship's RapidMiner AI Hub administration modules introduced this operational perspective and extended it to enterprise-level platform operations.

4.8 Methodology

The internship methodology followed a structured weekly progression. The learning process combined module-based study, practical engagement, weekly assessments, documentation, and a final assessment. The methodology can be understood through the following stages.

1. Module-based learning: Each week was assigned a defined module with a specific competency focus. The sequence moved from applications and use cases to data engineering, machine learning, advanced preparation, deployment and monitoring, and platform administration.

2. Hands-on practice: The learning pathway was practice-oriented, particularly in areas involving data access, transformation, machine learning, data preparation, and platform administration. Hands-on engagement was used to reinforce conceptual understanding.

3. Weekly assessments: Weekly learning was accompanied by assessment activities. These assessments provided checkpoints for understanding the material covered in each module and encouraged consistent progress across the eight-week schedule.

4. Project documentation: The internship experience and learning outcomes were consolidated through documentation. Preparing this report is part of that documentation process and provides a structured record of the internship scope, methods, tools, and outcomes.

5. Progressive integration: The modules were not treated as isolated topics. The sequence encouraged integration: use cases informed data requirements; data engineering supported machine learning; machine learning connected to deployment and monitoring; and the complete workflow was placed within a platform-administration context.

6. Credential validation and final assessment: The internship concluded with credential validation / internship grade-point assessment through the Final Assessment Test. This final stage provided an overall checkpoint for completion and learning validation.

4.9 Expected Outcomes

- Integrated understanding of the data-science lifecycle from problem identification to platform operation.
- Improved ability to recognize business problems that may be suitable for artificial intelligence and machine learning.
- Foundational competence in data access, loading, transformation, and calculation.
- Improved understanding of advanced data preparation and automation.
- Stronger conceptual and practical foundation in both basic and advanced machine-learning techniques.
- Awareness of deployment and monitoring as essential parts of the machine-learning lifecycle.
- Exposure to RapidMiner AI Hub configuration, management, and enterprise-level platform operations.

- Improved ability to connect academic computer-science concepts with structured industry-oriented workflows.

- Completion of the internship and associated credential validation process, including the issued virtual internship certificate.

- Enhanced employability through broader exposure to data-science concepts, tools, workflows, and professional terminology.

The internship certificate supplied with the report records successful completion of the eight-week Data Science Master Virtual Internship during June–August 2026. The certificate also identifies the candidate's Student ID and the officials associated with the internship.

4.10 Certificates of Completion and Communication Proof

EduSkills®
Nation Building Through Skills

Ref No: C17Y2026M091500473

Internship Offer Letter

Student Name	: Kushagra Pratap Singh
Stream	: B.Tech
Branch	: B.Tech(Computer Science and Engineering)
Institute Name	: IILM University, Greater Noida
Internship Domain	: Siemens Data Science Master
Internship Duration	: June 2026 - August 2026
AICTE Student ID	: STU6a0dfd7bdb6161779301755

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

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Nation Building Through Skills

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Build foundational machine learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
Week 7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.

Bibek Ranjan
Director, EduSkills

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Certificate details visible in the supplied certificate include: Certificate ID 4d1eb79183bf19fb6a43; Student ID STU6a0dfd7bdb6161779301755; internship title Data Science Master Virtual Internship; duration of eight weeks during June–August 2026; and support by Siemens. The certificate also displays an internship grade of O (Outstanding).

5. BIBLIOGRAPHY / REFERENCES

AICTE, Ministry of Education. Internship and student-development information associated with the AICTE–EduSkills internship framework. Program materials supplied for the present internship.

EduSkills. Data Science Master Virtual Internship. Internship modules, learning pathway, assessment structure, and certificate supplied by the student.

Siemens Industry Software (India) Pvt. Ltd. Data Science Master Virtual Internship support and associated program documentation supplied by the student.

RapidMiner. RapidMiner AI Hub documentation and platform-administration learning resources referenced by the internship module structure.

IILM University, School of Computer Science and Engineering. Internship Report Format updated 26-27. Institutional report-format document supplied for this report.

Certificate of Virtual Internship issued to Kushagra Pratap Singh. Data Science Master Virtual Internship, June–August 2026. Certificate supplied as part of the internship documentation.